

Neo-81 — Full Hardware Validation Checklist

Generated 2026-04-18. Updated 2026-04-28 to include all commits through cf931ab. All features implemented; none yet verified on hardware. Mark each test: ✔ PASS
✖ FAIL ⚠ PARTIAL

Pre-flight (do before any section)

- ☐ Latest firmware builds with 0 errors
- ☐ All host tests pass (`cmake -S App/Tests -B build-tests && cmake --build build-tests && ctest --test-dir build-tests`)
- ☐ Flash to board; power-cycle (USB unplug/replug — not just SWD reset)

Section 1 — Cursor rendering (P28)

Verifies `cursor_render()` refactor produces zero visible behaviour change across all 7 editors.

Main expression cursor

#	Action	Expected	Result
1	Power on, wait for home screen	Cursor blinks grey block on expression row	
2	Press 2ND	Cursor turns amber with ^ inside	
3	Press any function key (e.g. SIN)	Cursor resets to grey block	
4	Press ALPHA	Cursor turns green with A inside	
5	Press any letter key (e.g. A)	Letter inserted, cursor resets to grey	
6	Press STO→	Cursor turns green with A inside (STO-pending state)	
7	While STO-pending, press any letter	Stores to that variable, cursor resets to grey block	
8	Press 2ND+INS	Cursor changes to underscore style	
9	Press 2ND+INS again	Cursor returns to full-height block	

Y= editor

#	Action	Expected	Result
10	Press Y=	Cursor blinks on first equation field	
11	Press 2ND	Cursor turns amber with ^	
12	Press any key to resolve 2ND	Cursor resets	
13	Press 2ND+INS	Cursor changes to underscore style	
14	Press 2ND+INS again	Cursor returns to block	

RANGE editor

#	Action	Expected	Result
15	Press RANGE	Cursor blinks in Xmin field	
16	Press 2ND+INS	Cursor changes to underscore style	
17	Press 2ND+INS again	Cursor returns to block	

ZOOM FACTORS editor

#	Action	Expected	Result
18	ZOOM → Select Factors	Cursor blinks in XFact field	
19	Press 2ND+INS	Cursor changes to underscore style	
20	Press 2ND+INS again	Cursor returns to block	

Matrix editor — insert mode must NOT change cursor shape

#	Action	Expected	Result
21	MATRX → EDIT → select matrix → navigate to a cell	Cursor is full-height grey block	
22	Press 2ND+INS	Cursor shape does not change	
23	Press 2ND	Cursor turns amber with ^	
24	Press ALPHA	Cursor turns green with A	

PRGM name entry — insert mode must NOT change cursor shape

#	Action	Expected	Result
25	PRGM → EDIT → empty slot → ENTER	Cursor blinks in name field; full-height block	
26	Press 2ND+INS	Cursor shape does not change	

PRGM line editor

#	Action	Expected	Result
27	PRGM → EDIT → select a program	Cursor blinks on program line	
28	Press 2ND+INS	Cursor changes to underscore style	

29	Press 2ND+INS again	Cursor returns to block	
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P28 sign-off: all 29 pass → delete docs/p28_cursor_manual_tests.md , add row to docs/PROJECT_HISTORY.md .

Section 2 — Parametric graphing (P35h)

Validates *MODE*→*Param mode end-to-end*.

#	Test	Expected	Result
1	MODE → row 4 → select Param → press Y=	Y= screen shows $X_{1t}= / Y_{1t}=$ layout (not $Y_1=$)	
2	In param Y=, enter $\cos(T)$ for X_{1t} and $\sin(T)$ for Y_{1t} using T key	Equation accepted; T token inserted	
3	Set RANGE : Tmin=0, Tmax=6.28, Tstep=0.13, Xmin=-2, Xmax=2, Ymin=-2, Ymax=2 → press GRAPH	Circle rendered on graph canvas	
4	Press RANGE while in param mode	Exactly 9 fields shown: Tmin, Tmax, Tstep, Xmin, Xmax, Xscl, Ymin, Ymax, Yscl	
5	Press TRACE and use LEFT/RIGHT	Readout shows $T= / X= / Y=$ (not just $X= / Y=$)	
6	2nd+ON to save → power-cycle → enter param Y=	X_{1t}/Y_{1t} equations intact	

Section 3 — DRAW menu (P29h)

Validates *2nd+PRGM draw overlay commands*.

Setup: Ensure a Y= equation is entered (e.g. $Y_1=X$) and GRAPH has been rendered at least once so the graph canvas exists.

#	Test	Expected	Result
1	Press 2ND+PRGM	DRAW menu opens with 7 items: ClrDraw, Line(, PT-On(, PT-Off(, PT-Chg(, DrawF, Shade(. Digit shortcuts 1–7 work; UP/DOWN navigate; CLEAR exits.	
2	Type ClrDraw from expression buffer (via DRAW menu → item 1) → ENTER	Draw layer cleared; result row shows Done	
3	Type Line(0,0,5,5) → ENTER	Diagonal line drawn on graph canvas from (0,0) to (5,5)	
4	Type PT-On(2,3) → ENTER	Single pixel set at graph coordinates (2,3)	
5	Type PT-Off(2,3) → ENTER	That pixel cleared	
6	Type PT-Chg(2,3) → ENTER twice	Pixel toggles: on after first ENTER, off after second	
7	Type DrawF sin(X) → ENTER	Sine curve drawn as white overlay on graph canvas	
8	Type Shade(-1,1) → ENTER	Horizontal band between $y=-1$ and $y=1$ shaded on graph canvas (constant boundaries)	
9	After drawing, press ZOOM → Standard → return to graph	Draw layer content still visible (persists across re-render)	
10	Type ClrDraw → ENTER	All drawn content cleared from graph canvas	
11	Set window X[-5,5] Y[-5,5] → type Shade(sin(X),cos(X),2,-3,3) → ENTER	Shaded region appears only where $\sin(X) < \cos(X)$ within $X[-3,3]$; boundary curves drawn; resolution=2 visibly sparser than default	
12	Type Shade(X+1,X^3-8*X) → ENTER	Shading appears in columns where $X+1 < X^3-8X$; boundary curves drawn (guidebook p. 5-10 example)	
13	Type Shade(sin(X),cos(X),1,-1,1) → ENTER, then Shade(sin(X),cos(X),8,-1,1) → ENTER	Resolution 1 produces denser fill than resolution 8	

Section 4 — STAT menu (P30h)

Validates *statistics data entry, calculations, and graph plots*.

Setup: Before starting, open STAT → DATA → Edit and confirm the list is empty (or ClrStat first).

#	Test	Expected	Result
1	Press 2ND+MATRX	STAT menu opens with 3 tabs: CALC / DRAW / DATA	
2	DATA tab → Edit → enter 5 pairs: (1,3), (2,5), (3,7), (4,9), (5,11)	All 5 pairs accepted and visible in the editor	
3	CALC → 1:1-Var	Results: $n=5$, $\bar{x}=3$, $S_x \approx 1.5811$ (accept ± 0.001)	
4	CALC → 2:LinReg	Results: $a=2$, $b=1$, $r=1$; variables A=2 and B=1 set in calc engine (verify with A and B on home screen)	
5	DATA → 3:xSort	x-values sorted ascending; y-values reordered to match	
6	DATA → 4:ySort	y-values sorted ascending; x-values reordered to match	
7	Set RANGE: Xmin=0, Xmax=6, Ymin=0, Ymax=12 → DRAW → 2:Scatter	Scatter plot of data points visible on graph canvas	
8	DRAW → 3:xyLine	Points connected by lines on graph canvas	
9	DRAW → 1:Hist	Histogram bars visible on graph canvas	
10	2nd+ON to save → power-cycle → STAT → DATA → Edit	All 5 data pairs intact	
11	DATA → Edit → navigate to row 3 → press RIGHT to reach column 3 (row-select state)	> prefix appears before row 3 cursor; pressing INS/DEL applies to whole row	
12	With > cursor at row 3 → press 2ND+INS	New row (0,0) inserted before the current row 3; remaining rows shift down; total becomes 6 pairs	

13	With > cursor at a row → press DEL	That row is removed; remaining rows shift up	
14	From home screen, after running 1-Var with 5 pairs: type {x}{1} → ENTER	Result is the first x-value (1); type {y}{3} → ENTER: result is the third y-value (7); type {x}{6} → ENTER: DOMAIN ERR (out of bounds)	

Section 5 — VARS menu (P31h)

Validates all 5 tabs of the VARS key menu.

Setup: Run 1-Var (with the 5 stat pairs from Section 4) and LinReg before starting, so XY/LR/ Σ tabs have values. Ensure at least one matrix has non-trivial dimensions.

#	Test	Expected	Result
1	Press VARS	5-tab menu opens: XY / Σ / LR / DIM / RNG. LEFT/RIGHT navigate tabs; UP/DOWN navigate items; digit shortcuts 1–9/0 work; CLEAR exits	
2	RNG tab	Shows 10 items. Scroll indicators (↑↓) appear when scrolling past 7. Item 0:Tstep shortcut via 0 key inserts current Tstep value into expression	
3	XY tab → select 2: $\bar{x}$	Correct mean value (3) inserted into expression	
4	XY tab → select 3:Sx	Correct Sx value (≈ 1.5811) inserted	
5	DIM tab → select 1:Arow	Current row count of matrix [A] inserted	
6	DIM tab → select 2:Acoll	Current col count of matrix [A] inserted	
7	Press Y=, move cursor to Y ₁ = equation field, press VARS → XY → 2: $\bar{x}$	Value inserted into Y= equation (not home screen)	
8	LR tab → select 1:a	Regression coefficient a=2 inserted	
9	LR tab → select 4:RegEQ	String aX+b inserted into expression	
10	DIM tab → scroll to item 7:Dim{x} → select	Current stat list length (5, if 5 pairs were entered) inserted into expression; verify by pressing ENTER and confirming the value matches the pair count	

Section 6 — Y-VARS menu (P32h)

Validates 2nd+VARS equation reference and enable/disable tabs.

Setup: Ensure Y₁=X is entered and enabled. Enter Y₂=X² as well. In parametric mode, ensure X₁t=cos(T) / Y₁t=sin(T) is entered.

#	Test	Expected	Result
1	Press 2ND+VARS	3-tab menu opens: Y / ON / OFF. LEFT/RIGHT navigate tabs; UP/DOWN navigate items; digit shortcuts 1–5 work; CLEAR exits	
2	Y tab → select 1:Y ₁	String Y ₁ inserted into expression buffer	
3	Store 3→X; type Y ₁ (via Y-VARS) → ENTER	Result is 3 (evaluates Y ₁ =X at X=3)	
4	Press Y=; move cursor into Y ₁ equation field; press 2ND+VARS → Y → select 2:Y ₂	Y ₂ inserted into the Y= equation field (not home screen)	
5	OFF tab → 1:All-Off → press Y=	All equations show – (disabled); no = visible	
6	ON tab → 2:Y ₁ -On → press Y=	Y ₁ shows = (enabled); Y ₂ -Y ₄ remain –	
7	Y tab → press 1 directly (no UP/DOWN first)	Y ₁ inserted immediately — digit shortcut works without navigating	
8	In Func mode: Y tab shows exactly 4 items (Y ₁ -Y ₄); in Param mode: Y tab shows 10 items (Y ₁ -Y ₄ then X ₁ t, Y ₁ t, X ₂ t, Y ₂ t, X ₃ t, Y ₃ t) with scroll indicators (↑↓) appearing once past item 7	Switch to Param via MODE; re-open Y-VARS → Y tab; confirm 10 items and scroll	
9	In Param mode: Y tab → press 6 (digit shortcut for item 6)	X ₁ t string inserted immediately	
10	In Param mode: Y tab → scroll down to item 8:Y ₂ t → ENTER	Y ₂ t inserted into expression buffer	
11	In Param mode: ON tab	8 items visible: 1:All-0n, 2:Y ₁ -0n, 3:Y ₂ -0n, 4:Y ₃ -0n, 5:Y ₄ -0n, 6:X ₁ t-0n, 7:X ₂ t-0n, 8:X ₃ t-0n	
12	In Param mode: OFF tab → select 6:X ₁ t-Off → press Y=	X ₁ t/Y ₁ t pair shows – (disabled); other pairs unaffected	

Section 7 — PRGM system (P10)

50 tests + 5 new. Hardware: STM32F429I-DISC1. Run after flashing latest build. Mark  /  / . See Notes section at end for context on individual tests.

Menu & Navigation

#	Test	Steps	Expected	Result
T01	Open PRGM menu	From home, press PRGM	PRGM menu opens: EXEC EDIT ERASE tab bar, EXEC tab yellow. All 37 slots in format N:PrgmN. Named slots show N:PrgmN NAME in second column.	
T02	Tab navigation with wrap	With PRGM open, press RIGHT twice	Tab advances EXEC→EDIT→ERASE. LEFT at EXEC wraps to ERASE; RIGHT at ERASE wraps to EXEC. All 3 tabs show all 37 slots.	
T03	Scroll indicators	EXEC or EDIT tab → press DOWN past slot 7	↓ amber indicator at bottom when more items below; ↑ at top once scrolled. Disappear at boundaries.	
T04	Close menu	With PRGM open, press CLEAR	PRGM menu closes; home screen returns; expression unchanged	
T05	Open PRGM from MATH menu	MATH key → then press PRGM	MATH closes; PRGM opens on EXEC tab; no display corruption	

Name Entry & Program Creation

#	Test	Steps	Expected	Result
T06	Create new program with name	PRGM → EDIT → empty slot → ENTER → type TEST → ENTER	Name-entry screen appears with cursor in alpha mode (green A). On ENTER, editor opens with title Prgm1 TEST and blank : line.	
T06a	Name-entry LEFT/RIGHT cursor	On name-entry with ABCD typed, press LEFT twice → type X	X inserted: field reads ABXCD. RIGHT RIGHT moves to end; DEL removes last char.	
T07	Create program — skip name	EDIT → empty slot → ENTER → immediately ENTER again	Editor opens with title PrgmN (no user name). Body editable.	
T08	Digits allowed in name	EDIT → empty slot → ENTER → type A1B2 (ALPHA for letters, plain for digits)	A, 1, B, 2 all accepted. Digits typed without ALPHA.	
T08a	DOWN from name-entry opens body	Name-entry with MYTEST typed → press DOWN	Editor body opens; title shows PrgmN MYTEST. Pressing UP from line 0 at col 0 returns to name-entry with name intact.	
T09	Name-entry DEL	Type a few letters → press DEL	Last character removed on first DEL press. ALPHA mode re-engages automatically.	
T09d	ALPHA_LOCK in name entry	From name-entry, press 2ND+ALPHA to engage ALPHA_LOCK	Cursor green A. Typing letters inserts into name field. Pressing ALPHA once exits ALPHA_LOCK.	
T09b	ENTER works on first press	Type name → press ENTER exactly once	Editor opens on first ENTER; second press not needed	
T09c	CLEAR works on first press	Type letters in name entry → press CLEAR	Returns to PRGM EDIT tab on first CLEAR; no name or body saved	
T10	Cancel name entry (empty)	From name-entry (no letters typed) → CLEAR	Returns to PRGM EDIT tab	
T11	Open existing program	EDIT tab → navigate to named slot → ENTER	Editor opens directly (no name-entry screen). Title shows name.	
T11b	EDIT tab digit shortcut	PRGM → EDIT tab → press 1 without UP/DOWN	Editor for slot 1 opens immediately	

Editor

#	Test	Steps	Expected	Result
T12	Character input	Open any program → type A+1	Line shows :A+1; cursor advances	
T12b	Insert mode off by default	Open program → navigate LEFT → type a character (no INS pressed)	Character overwrites (no push-right). INS key toggles insert mode.	
T13	CTL menu inserts keyword	In editor → PRGM → 3:If (or press 3)	CTL closes; line shows :If. Menu has exactly 8 items: Lbl, Goto, If, IS> (, DS<(, Pause, End, Stop.	
T14	I/O menu inserts keyword	In editor → PRGM → LEFT/RIGHT to I/O tab → 1:Disp (or press 1)	Line shows :Disp. Menu has exactly 5 items: Disp, Input, DispHome, DispGraph, ClrHome.	
T15	Multi-line scroll	In editor → press ENTER 6+ times	↓ indicator on last visible row; ↑ appears after scrolling. Title stays fixed.	
T16	Erase a program	PRGM → ERASE tab → navigate to named slot → ENTER	Confirmation dialog: 1:Do not erase / 2:Erase. Press 2 — slot cleared immediately (also test digit shortcuts on the confirmation dialog).	
T16b	ERASE shows all 37 slots	Open PRGM → ERASE tab	All 37 slots visible (named and unnamed). Named show N:PrgmN NAME; unnamed show N:PrgmN.	

Persistence

#	Test	Steps	Expected	Result
T17	Programs survive power cycle	Create program SAVE in slot 1 with Disp "OK" → 2nd+ON → USB unplug/replug	1:Prgm1 SAVE in EXEC/EDIT lists; body intact	

Executor — Basic Execution

#	Test	Steps	Expected	Result
T18	Run Disp program	Slot 1 has HELLO with body Disp "HELLO" → EXEC → slot 1 → ENTER → ENTER	HELLO in history (left-aligned, grey). Done as result row (right-aligned, white).	
T19	Run empty slot	EXEC → slot with no body → ENTER → ENTER	Done appears; no error or lockup	
T20	Expression evaluation and ANS	Program: 2+2 then Disp ANS → run	4 in history output; ANS=4 after	
T21	Input and variable store	Program: Input A then Disp A → run → type 7 → ENTER	? prompt shown; after ENTER: 7 displayed; A=7	
T22	If single-line skip	Program: 0→A / If A=1 / Disp "YES" / Disp "DONE" → run	YES does not appear; DONE appears	
T23	EXEC number-key shortcut	PRGM → EXEC tab → press 1 (no UP/DOWN)	prgm1 (or prgmNAME) inserted in expression buffer	
T24	CLEAR aborts execution	Program: Lbl A / Disp "X" / Goto A → run → press CLEAR while running	Execution stops; home screen returns; no lockup	

Executor — Advanced Control Flow

#	Test	Steps	Expected	Result
T25	Lbl/Goto label entry	In editor → CTL → Lbl → type A → try to type a second letter	(Note: single-char enforcement is a known intentional constraint; multi-char throws error on execution, not on entry)	

T26	Disp string — left alignment	Program: Disp "HELLO" → run	HELLO left-aligned, grey	
T27	Disp variable — right alignment	Program: 5→A / Disp A → run	5 right-aligned, white	
T28	Goto/Lbl jump	Program: Goto END / Disp "SKIP" / Lbl END / Disp "DONE" → run	SKIP absent; DONE appears once	
T29	Pause halts and resumes	Program: Disp "WAIT" / Pause / Disp "RESUMED" → run	Halts after WAIT ; ENTER resumes; RESUMED appears	
T30	Stop terminates early	Program: Disp "A" / Stop / Disp "B" → run	A appears; B does not	
T31	IS>(increment and skip	Program: 1→I / IS>(I,2) / Disp "SKP" / Disp I → run	I=1→2; 2 is not >2 so SKP NOT skipped; I shows 2	
T32	DS<(decrement and skip	Program: 3→I / DS<(I,3) / Disp "SKP" / Disp I → run	I=3→2; 2<3 so SKP IS skipped; I shows 2	
T33	Subroutine auto-return	Slot 2: Disp "SUB" . Slot 1: Disp "MAIN" / prgm2 / Disp "BACK" (insert via PRGM→EXEC) → run slot 1	History: MAIN , SUB , BACK in order	
T33b	EXEC sub-menu tab	In editor → PRGM → RIGHT → RIGHT	EXEC tab (third tab) highlighted. Shows all 37 slots. Select slot 2 → :prgm2 inserted. LEFT from CTL wraps to EXEC.	
T34	Nested subroutine (2 deep)	Slot 3: Disp "DEEP" . Slot 2: prgm3 / Disp "MID" . Slot 1: prgm2 / Disp "TOP" → run slot 1	History: DEEP , MID , TOP in order	

Executor — I/O Commands

#	Test	Steps	Expected	Result
T35	prgmNAME execution (named)	EXEC → named slot (e.g. 1:Prgm1 TEST) → ENTER → ENTER	prgmTEST in expression (grey, left-aligned); after run: Done result row	
T35b	prgmNAME execution (unnamed)	EXEC → unnamed slot (e.g. 3:Prgm3) → ENTER → ENTER	prgm3 in expression; Done result row	
T36	ClrHome clears history	Program: Disp "LINE1" / Disp "LINE2" / ClrHome / Disp "AFTER" → run	After ClrHome, LINE1 and LINE2 gone; only AFTER visible	
T37	DispGraph switches to graph view	Ensure Y ₁ =X. Program: DispGraph / Pause / DispHome → run	Graph canvas appears (no lockup); ENTER at Pause switches back to home screen	

Executor — End command

#	Test	Steps	Expected	Result
T38	End terminates at top level	Program: Disp "A" / End / Disp "B" → run	A appears; B does not; Done shown (same behaviour as Stop)	
T39	End returns from subroutine	Slot 2: Disp "SUB" / End . Slot 1: Disp "MAIN" / prgm2 / Disp "BACK" → run slot 1	History: MAIN , SUB , BACK in order — End in subroutine returns to caller, does not terminate the parent	

PRGM MODE sub-menu

#	Test	Steps	Expected	Result
T40	PRGM MODE opens NUMBER tab	In editor → press MODE key	PRGM MODE sub-menu opens with NUMBER tab active; 7 items: 1:Norm , 2:Sci , 3:Eng , 4:Fix , 5:Float , 6:Rad , 7:Deg ; CLEAR exits	
T41	PRGM MODE NUMBER item inserts keyword	Select 2:Sci	:Sci inserted into current program line	
T42	PRGM MODE GRAPH tab	From NUMBER tab, press RIGHT	GRAPH tab active; 10 items: 1:Function – 9:Rect with 0: (10th) requiring scroll; LEFT/RIGHT tab wrap works	
T43	PRGM MODE executes notation change	Slot 1: Sci . Run slot 1 from home. Type 123 → ENTER	Result shows 1.23E2 — Sci mode active; confirm by re-running Norm program and checking 123 ENTER → 123	

Editor Alpha Behaviour

#	Test	Steps	Expected	Result
T41	Cursor blinks in editor	Open any program; observe for ~2 seconds	Cursor blinks at ~530 ms interval on current line	
T42	2ND mode in editor	In editor → press 2ND	Cursor turns amber with ^ ; resolves on next non-modifier key	
T43	ALPHA single in editor	In editor → press ALPHA once	Cursor green A ; one letter inserted; cursor resets to white. Second ALPHA press cancels ALPHA mode.	
T43b	ALPHA_LOCK in editor	In editor → press 2ND+ALPHA	Cursor green A ; multiple letters inserted into program body; ALPHA press once cancels; digits work without exiting ALPHA_LOCK	
T44	Cursor blinks in name entry	Open name-entry screen; observe for ~2 seconds	Cursor blinks at ~530 ms interval	

Editor Body Behaviour

#	Test	Steps	Expected	Result
T45	Body-only slot opens directly	Editor → type content → CLEAR at name-entry (skip name). Reopen EDIT → same slot → ENTER	Editor opens directly (no name-entry screen); body preserved	
T46	Tab wrap in sub-menus	In editor → PRGM → CTL tab. Press LEFT.	Wraps to EXEC. LEFT→I/O; LEFT→CTL. RIGHT: CTL→I/O→EXEC→CTL	

T46b	EXEC digit and ALPHA+letter shortcuts	On EXEC sub-menu → press 2	:prgm2 inserted immediately. Then EXEC tab again → ALPHA+A → :prgmA inserted (slot 10).	
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Notes (PRGM section)

- All tab/item highlights are **yellow** (0xFFFF00). Scroll indicators are **amber** (0xFFAA00). Amber on a tab or item cursor (not an arrow) is a regression.
- T17 requires USB unplug/replug (full power cycle), not just SWD reset.
- T33/T34: max call stack depth is 4; exceeding it makes the prgm call a no-op (no crash). No explicit Return command — subroutines auto-return at end of body.
- T24 CLEAR abort: a few X rows may appear before CLEAR is processed (queue latency) — acceptable.
- CTL menu must have exactly **8 items**. Then, Else, While, For(, Return, prgm must not appear.
- I/O menu must have exactly **5 items**. Prompt, Output(, Menu(must not appear.
- T25 label enforcement: single-char limit is enforced at entry in the editor (typing a second char is ignored). This is intentional. If you want to change this to allow multi-char labels with a runtime error, add it to "Next session priorities" in CLAUDE.md.
- T40–T43: PRGM MODE NUMBER/GRAPH tab numbers listed above; verify via docs/PRGM_COMMANDS.md for full item list.

Section 8 — MODE screen — display and graph modes

Validates MODE rows 0, 5, 6, 8 and the free-roaming graph cursor introduced with MODE_GRAPH_FREE_CURSOR.

Setup: Start in default state (Normal notation, Func mode, Connected, Sequential, Rect display). Use 2nd+0N → power-cycle to verify persistence tests.

Display notation (Sci / Eng / Norm)

#	Test	Expected	Result
1	Press MODE → inspect row 0	Three options shown: Norm Sci Eng ; current selection highlighted	
2	Select Sci → ENTER → return to home → type 12345 → ENTER	Result shows 1.2345E4 (exactly one digit before decimal point)	
3	In Sci mode, type 0.00123 → ENTER	Result shows 1.23E-3	
4	Select Eng → ENTER → type 12345 → ENTER	Result shows 12.345E3 (exponent is a multiple of 3)	
5	In Eng mode, type 0.00123 → ENTER	Result shows 1.23E-3	
6	Select Norm → ENTER → type 0.0005 → ENTER	Result shows 5E-4 (auto-sci threshold: values < 0.001 display in scientific notation)	
7	In Norm mode, type 0.001 → ENTER	Result shows 0.001 (threshold is strictly less than 0.001, so 0.001 stays in Norm)	
8	Set Sci → 2nd+0N → power-cycle → check home screen with 100 → ENTER	Result shows 1E2 — notation mode persisted	

Free-roaming graph cursor

Setup: Ensure $Y_1=\sin(X)$ is entered and enabled. Set RANGE to ZTrig preset (ZOOM → ZTrig).

#	Test	Expected	Result
9	Press GRAPH from home screen	Blinking white crosshair appears at screen centre; X= and Y= labels below canvas show current math coordinates	
10	Press LEFT, RIGHT, UP, DOWN	Crosshair moves one pixel per press; X= and Y= update with each step; cursor remains visible immediately after each press (blink timer resets)	
11	From free-cursor position, press TRACE	Cursor snaps to nearest active equation at the current X position; crosshair turns to trace style (green); X= / Y= show on-curve values	
12	While in TRACE mode, press TRACE again	Cursor re-snaps to X=x_mid on the same equation — does NOT exit trace mode; no visible glitch	
13	While in TRACE mode, press GRAPH	Graph re-renders; blinking white free cursor appears at screen centre	
14	While in TRACE mode, press a digit key (e.g. 5)	Trace exits; 5 appears in expression buffer on home screen	
15	From Y= screen, press GRAPH	Blinking free cursor appears (not a static render)	
16	From RANGE screen, press GRAPH	Blinking free cursor appears	
17	From ZOOM menu, press GRAPH	Blinking free cursor appears	
18	Navigate to home screen or any menu after graphing	Cursor blink timer stops — no blinking animation occurs when graph canvas is not the active screen	

Connected / Dot plot mode (P33h)

#	Test	Expected	Result
19	Press MODE → row 5 shows Connected Dot	Both options visible; current selection highlighted	
20	Select Dot → ENTER → enter $Y_1=\sin(X)$ → set ZTrig preset → GRAPH	Only individual pixels plotted; no line segments connecting them	
21	Select Connected → ENTER → re-GRAPH (or press GRAPH again)	Interpolated line segments connect the plotted pixels	
22	While in Dot mode, select MODE row 4 Param → ENTER	Y= layout switches to X_1t/Y_1t (row index bug verified fixed — no mis-routing to Connected/Dot row)	
23	Set Dot → 2nd+0N → power-cycle → check MODE row 5	Dot remains selected	

Sequential / Simultaneous plot mode (P38h)

Setup: Enter $Y_1=X$ and $Y_2=X^2$ in Func mode. Set RANGE: Xmin=-5, Xmax=5, Ymin=-5, Ymax=25.

#	Test	Expected	Result
24	Press MODE → row 6 shows Sequential Simul	Both options visible	
25	Select Simul → ENTER → GRAPH	Both Y_1 and Y_2 advance across the screen simultaneously — visible interleaving as the graph draws	
26	Select Sequential → ENTER → GRAPH	$Y_1=X$ plots completely first, then $Y_2=X^2$ plots	
27	Set Simul → 2nd+ON → power-cycle → check MODE row 6	Simul remains selected	

Polar coordinate display (P40h)

Setup: Return to Rect mode and Func mode. Ensure $Y_1=X^2$ is entered.

#	Test	Expected	Result
28	Press MODE → row 8 shows Rect Pol	Both options visible	
29	Select Pol → ENTER → GRAPH with $Y_1=X^2$ → move free cursor	Readout below canvas shows $R=$ and $\theta=$ instead of $X=$ / $Y=$	
30	In Pol display mode, press TRACE and move LEFT/RIGHT	Trace readout shows $R=$ / $\theta=$ for each step	
31	Select Rect → ENTER → GRAPH	Readout reverts to $X=$ / $Y=$	
32	Set Pol → 2nd+ON → power-cycle → check MODE row 8	Pol remains selected	
33	On home screen (Rad angle mode), type $R>P(-1,\theta)$ → ENTER	Result is 1; then type θ → ENTER : result is 3.14159...	
34	Type $P>R(1,\theta)$ → ENTER	Result is 1; then type Y → ENTER : result is 0	
35	Type 3 → STO → θ → ENTER → type θ → ENTER	θ variable stores 3; second ENTER returns 3	

Section 9 — MATH extensions (HYP tab + nDeriv + expression fixes)

Validates hyperbolic functions, numerical derivative, and auto-close trailing parenthesis.

Setup: Ensure angle mode is Rad.

MATH HYP tab

#	Test	Expected	Result
1	Press MATH → navigate to HYP tab	HYP tab shows 6 items: sinh (, cosh (, tanh (, asinh (, acosh (, atanh (	
2	Select 1: sinh (→ type 0 → ENTER	Result is 0	
3	Select 2: cosh (→ type 0 → ENTER	Result is 1	
4	Select 3: tanh (→ type 0 → ENTER	Result is 0	
5	Select 4: asinh (→ type 1 → ENTER	Result ≈ 0.881374	
6	Select 5: acosh (→ type 1 → ENTER	Result is 0	
7	Select 5: acosh (→ type 0 → ENTER	DOMAIN ERR (acosh domain is $x \geq 1$)	
8	Select 6: atanh (→ type 0 → ENTER	Result is 0	
9	Select 6: atanh (→ type 1 → ENTER	DOMAIN ERR (atanh domain is	x
10	Type sinh (1) directly in expression buffer without menu	Result ≈ 1.1752 (confirms keyword tokenizer matches sinh before sin)	

nDeriv(

#	Test	Expected	Result
11	Type nDeriv ($X^2,X,3$) → ENTER	Result ≈ 6 (derivative of X^2 at $X=3$ is $2X=6$; tolerance ± 0.01)	
12	Type nDeriv (sin (X), $X,0$) → ENTER	Result ≈ 1 (derivative of sin at 0 is cos (0)=1; tolerance ± 0.01)	
13	Type nDeriv ($X^3,X,2$) → ENTER	Result ≈ 12 (derivative of X^3 at $X=2$ is $3X^2=12$; tolerance ± 0.01)	

Auto-close trailing parenthesis

#	Test	Expected	Result
14	Type sin (1 (no closing)) → ENTER	Result ≈ 0.841471 — unmatched (is auto-closed; no SYNTAX ERR	
15	Type (1+2 → ENTER	Result is 3 — leading unmatched (auto-closed	
16	Type sin (cos (θ → ENTER	Result ≈ 0.841471 — two unmatched (auto-closed in order	

Section 10 — Matrix row operations (MATRX MATH menu)

Validates **rowSwap**, **Row+**, ***Row**, ***Row+** added to **MATRX MATH** menu.

Setup: Set matrix [A] to a 2x2 matrix with values [[1,2],[3,4]]. (**MATRX** → **EDIT** → [A] → set 2 rows, 2 cols → enter values.)

#	Test	Expected	Result
1	Press MATRX	MATRX menu opens; navigate to MATH tab	MATH tab shows 6 items: 1: det (, 2: T , 3: rowSwap (, 4: Row+ (, 5: *Row (, 6: *Row+ (
2	Select 1: det (→ type [A]) → ENTER	Result is -2 (det of [[1,2],[3,4]] = $1 \times 4 - 2 \times 3 = -2$)	
3	Select 2: T → type [A]) → ENTER	Result is [[1,3],[2,4]] — transpose of [A]	
4	Select 3: rowSwap (→ type [A],1,2) → ENTER	Result is [[3,4],[1,2]] — rows 1 and 2 swapped	
5	Select 4: Row+ (→ type [A],1,2) → ENTER	Result is [[1,2],[4,6]] — row 1 added to row 2	

6	Select 5:*Row(→ type 2, [A],1) → ENTER	Result is [[2,4],[3,4]] — row 1 multiplied by scalar 2	
7	Select 6:*Row+(→ type 2, [A],1,2) → ENTER	Result is [[1,2],[5,8]] — 2×row 1 added to row 2	
8	All row-op results: original [A] unchanged after evaluating expressions	Press [A] → ENTER: still [[1,2],[3,4]] (row-ops return new matrix, do not modify in place)	

Section 11 — RESET menu

Validates 2nd++ (2nd then +) RESET MEMORY confirmation screen per guidebook p. 1-28.

Setup: Ensure some data exists before testing reset: enter a stat data pair, name a program, store a value to variable A.

#	Test	Expected	Result
1	Press 2ND then + from home screen	RESET MEMORY screen appears with title MEMORY , stat point count (Sts: N pts), program count (Pgm: N stored), and choices 1:No / 2:Reset	
2	On RESET screen: press 1 or CLEAR	RESET screen closes; returns to previous screen; no data changed	
3	Stat count display	Open RESET screen when 5 stat pairs are entered	Sts: 5 pts shown
4	Program count display	Open RESET screen when 2 programs are named	Pgm: 2 stored shown
5	Press 2 to confirm reset	RESET executes; home screen shows Mem cleared ; ANS=0	
6	After reset: verify stat cleared	STAT → DATA → Edit	List is empty
7	After reset: verify variables cleared	Type A → ENTER	Result is 0
8	After reset: verify Y= cleared	Press Y=	All equations show blank / disabled
9	After reset: verify RANGE defaults	Press RANGE	Xmin=-10, Xmax=10, Xscl=1, Ymin=-10, Ymax=10, Yscl=1
10	2ND++ accessible from non-home screens	Open STAT menu → press 2ND then +	RESET screen appears (TOKEN_RESET fires from any mode)

Summary

Section	Feature	Tests	Result
1	Cursor rendering (P28)	29	
2	Parametric graphing (P35h)	6	
3	DRAW menu (P29h)	13	
4	STAT menu (P30h)	14	
5	VARS menu (P31h)	10	
6	Y-VARS menu (P32h)	12	
7	PRGM system (P10)	55	
8	MODE screen (notation + graph modes + free cursor)	35	
9	MATH extensions (HYP + nDeriv + auto-close)	16	
10	Matrix row operations	8	
11	RESET menu	10	
Total		208	

Sign-off

When all sections pass:

- 1. Delete docs/p28_cursor_manual_tests.md (P28 complete)
- 2. Delete docs/prgm_manual_tests.md (P10 complete)
- 3. Add a Resolved Items row to docs/PROJECT_HISTORY.md for each section
- 4. Remove completed hardware items from "Next session priorities" in CLAUDE.md